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Started	Wednesday, 21 October 2026, 9:25 PM
Completed	Wednesday, 21 October 2026, 9:31 PM
Duration	5 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Bubble Sort
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Dividing the list into halves
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data in a single pass
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **5**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Deleting data from a list
- c. Inserting data into a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Heap Sort
- b. Merge Sort
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It divides the list into sublists
- c. It recursively sorts subproblems
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort

b. Selection sort

c. Merge sort

d. Heap sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Merging

c. Complexity

d. Sorting

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them

b. By comparing elements sequentially

c. By sorting data in a single pass

d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. sorted()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It only works with integer arrays
- c. It requires external libraries
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. sort()
- c. sorted()
- d. arrange()

The correct answer is: sort()